

An Integrated Approach to Knowledge and Prediction Modeling of Breast Cancer Metastasis Using Gene Regulatory Networks

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Abstract—Understanding why only some breast cancers become metastatic, and predicting metastatic risks at earlier cancer stages, are two major goals of breast cancer research. These goals are clearly connected and synergistic, but are rarely integrated within a single study. Knowledge discovery-oriented research has identified critical biological pathways related to metastasis, but the interplay between these pathways remains elusive, resulting in models that lack prediction accuracy. Conversely, complex machine learning models achieve high prediction accuracy without detailed biological knowledge, making the explanation challenging. Here, we propose a novel computational framework that addresses both goals simultaneously. To support knowledge discovery, we construct gene regulatory networks (GRNs) to model the cellular states of metastatic and non-metastatic patients. To enable explainable metastasis prediction, we introduce a dysregulation score based on the GRN models. Experimental results demonstrate that our method not only identified significant metastasis-associated GRN changes, but also revealed the loss of co-regulation among key biological processes in metastatic patients. Leveraging the dysregulation score, our model-free classifier outperformed complex machine learning models under rigorous evaluation. This work bridges a significant gap between knowledge discovery and accurate, explainable prediction by employing carefully designed knowledge models and knowledge-based prediction, with potential applicability in other disease contexts.

Index Terms—Gene regulatory networks, breast cancer metastasis, knowledge models, prediction models.

I. INTRODUCTION

BREAST cancer metastasis remains a substantial challenge in oncology due to its profound impact on patient prognosis

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and survival. The spread of cancer cells from the primary tumor to distant tissue locations significantly complicates effective treatment and often leads to incurable outcomes, accounting for more than 90% of cancer-related deaths [1]. Understanding and predicting metastasis in its incipient stages is critical for prompt interventions and customized therapeutic strategies [2].

The mechanism of metastasis is the most poorly understood process within the life cycle of cancer. According to somatic mutation theory, driver mutations in oncogenes turn healthy cells into abnormal tumor cells. Subsequent tumor cell populations go through an evolutionary selection process and acquire increased proliferation and immune resistance capabilities through cascading mutations, which ultimately leads to metastasis [3], [4], [5]. Although somatic mutation theory has been prevalent for nearly half a century as a guiding principle in cancer research, it has proved inadequate to explain many confounding observations made possible by advances in sequencing technology. Such observations include the lack of consistent association between driver mutations and disease progression, and phenotypic cell heterogeneity within the tumor microenvironment. It is now abundantly clear that cancer must be viewed through the lens of not individual genes and their mutations, but of gene regulatory programs and their rewiring [6]. To this end, dysregulation of transcription factors in vital pathways such as those governing the epithelial-mesenchymal transition (EMT) and cell dormancy has been studied to understand their contributions to metastatic behavior [7], [8], [9]. Furthermore, computational methods that integrate large-scale omics have provided valuable insight and helped identify novel pathways and sub-networks as potentially involved in disease progression [10], [11], [12], [13]. Despite these advances and the multitude of generated data, the interplay between regulatory factors and pathways is not adequately understood, hindering the development of robust knowledge-based translational models that can accurately describe metastasis.

While unraveling the driving mechanisms of metastasis has proven to be a daunting task, research into its early prediction has seen significant progress. Data-driven machine learning methods have shown immense potential in leveraging the growing repository of -omics data without the need to incorporate knowledge of the inner workings of the cell. These models capitalize on a variety of data sources, including genomic, transcriptomic, and epigenomic datasets, to predict the likelihood and behavior of metastasis [14], [15], [16]. Models trained on genotype data

initially focused on individual genes or functionally related groups of genes as features, with limited success [17], [18], [19]. Building on these foundational approaches, more sophisticated models have emerged, taking advantage of advanced machine learning methods [20], [21], [22] and knowledge graphs such as protein/gene networks or functional pathways [23], [24], [25], [26], [27] to extract and utilize high-level features. These approaches have been shown to improve accuracy by capturing the intricate interconnections and interdependencies present in biological systems.

Despite their impressive predictive capabilities, a significant drawback of machine learning models is their lack of explainability; they are often opaque and provide limited insight into biological processes. Although post-hoc analysis of these models can offer some gene- or pathway-level insights [28], translating data into meaningful biological knowledge poses a challenge. Consequently, comprehensive knowledge models, such as transcriptional regulatory networks, cannot be easily integrated into these machine learning frameworks. This gap highlights the ongoing challenge of aligning computational advances with tangible biological insights which ultimately limits the direct application of these predictions in clinical settings.

In this study, we propose the use of gene regulatory networks (GRNs) to address the limitations inherent in current methodologies and to bridge the divide between biological knowledge and predictive modeling. This approach involves constructing GRNs as knowledge models [29], [30] and introducing a novel GRN-based metric called the dysregulation score to predict metastasis. These knowledge models help pinpoint network changes associated with metastasis, enabling detailed insights into gene-level and pathway-level interactions disrupted during metastasis. Furthermore, a simple knowledge-based classifier based on the dysregulation score achieved remarkably high prediction accuracy compared to sophisticated, state-of-the-art classifiers. We believe that this provides a framework to achieve biologically explainable predictions, which can help identify potential therapeutic strategies. Our work bridges a significant gap between knowledge discovery and outcome classification in the study of breast cancer metastasis and has the potential to be applied to other complex diseases.

II. MATERIALS AND METHODS

A. Overview

Our methodology consists of two major segments, the first of which revolves around building knowledge representation models that accurately represent the state of the system for a given disease context. Gene regulatory networks represent regulatory relationships between genes through directed edges. Patient samples were divided into metastasized and non-metastasized classes, and for each class, regression models were trained on gene expression levels. These models served as the basis for constructing gene-regulatory networks that include genes and transcriptional regulators, which in turn were consolidated into co-regulatory networks for exploring the relations between transcription factors (TFs) in terms of their shared regulatory targets. The disruption of regulatory programs is reflected in the

topological differences between networks derived from metastatic and non-metastatic samples. In the second part, regression models were repurposed to produce a classifier that can utilize network differences to classify an unknown sample. The aforementioned models can be used to estimate the expressions of genes from the expressions of the TFs. The estimated expressions, in conjunction with the observed expression, served as input for a simple yet intuitive classification strategy that requires no additional learning or parameter tuning. An overview of the methodology is presented in Fig. 1. Finally, a comprehensive evaluation of classification performance suggested that our approach can outperform state-of-the-art classification algorithms trained on the same dataset while being substantially more explainable.

B. Datasets

The primary gene expression dataset used for this work is a preprocessed compilation of 1,616 patient samples from 12 separate breast cancer studies in the Amsterdam Classification Evaluation Suite (ACES) [31]. A total of 12,750 genes, 1,010 of which are TFs [32], were profiled using the Affymetrix U133 A platform, followed by duplicate removal, quality check, outlier removal, and cross-study normalization. Metastasis occurred within five years of initial diagnosis in 455 patients, and the remaining 1,161 were metastasis-free for at least five years. We define the following terms:

V, T	: set of genes and TFs, $T \subset V$
$m, N = V , t = T $	: # of samples, genes, and TFs
$Y = \{y_{k,j}\} \in \mathbb{R}^{m \times N}$	: gene expression matrix
$X = \{x_{k,i}\} \in \mathbb{R}^{m \times t}$	: TF expression matrix
$Y_{k,:} \in \mathbb{R}^{1 \times N}$	: expr. of genes in the k-th sample
$X_{k,:} \in \mathbb{R}^{1 \times t}$	: expr. of TFs in the k-th sample
$c_k \in [0, 1]$	: class label of the k-th sample
$S^+ = \{k \in [m] : c_k = 1\}$	: set of metastatic samples
$S^- = \{k \in [m] : c_k = 0\}$	: set of non-metastatic samples

Two additional expression datasets, NKI [18] and METABRIC [33] were used to validate the predictive performance of our method. The NKI dataset consists of expression profiles for 11,658 genes (907 TFs) for 295 patient samples (217 non-metastatic and 78 metastatic). The METABRIC dataset contains expression levels of 20,182 genes (1,531 TFs). For this dataset, we considered no disease progression within 5 years of initial diagnosis (1,341 patient samples) as *good outcomes*, while recurrence events within 5 years of initial diagnosis (495 patient samples) were considered to be *poor outcomes*. Both datasets were used as an independent test set for models trained on the primary dataset. They were also used in a cross-validation setting to train and test our model alongside baselines. Entrez gene IDs were used to map gene entries to the ones in the ACES dataset. After mapping, the NKI and ACES datasets have 10,398 genes and 820 TFs in common,

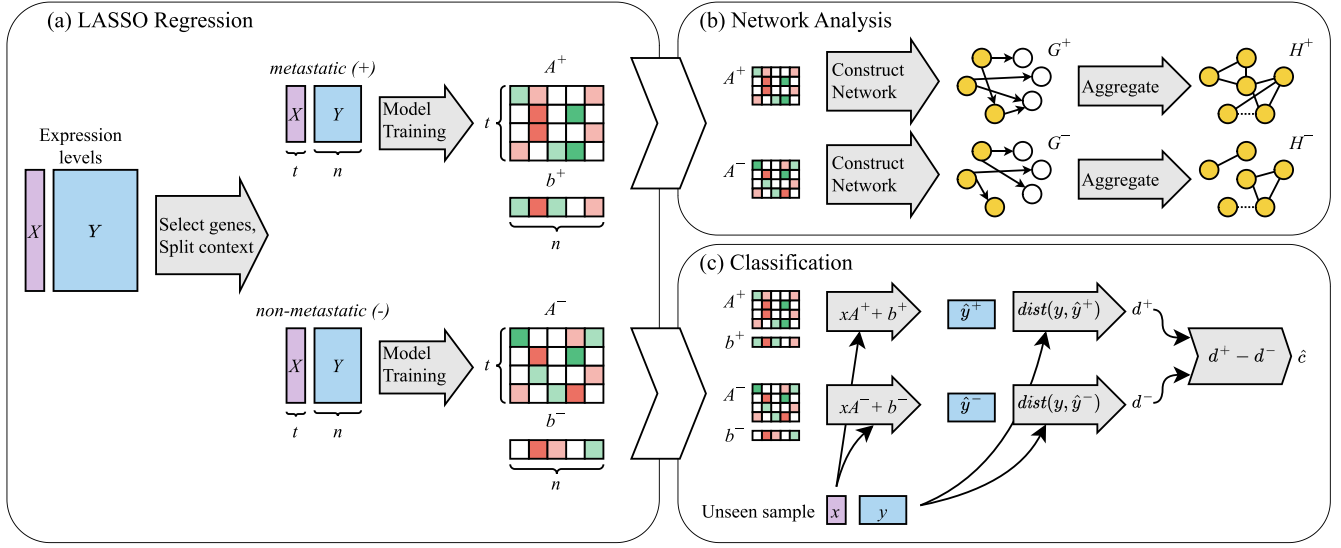


Fig. 1. Overview of the methodology. (a) Gene expression profiles of patient samples are split into two subsets based on the class labels. Each subset is used to train LASSO regression models. The coefficients and biases trained from the metastatic and non-metastatic samples are denoted (A^+, b^+) and (A^-, b^-) , respectively. (b) Each set of coefficients is used to infer a gene regulatory network. The network-level properties of the two GRNs are compared. Each GRN is used as the input in constructing a corresponding TF-TF co-regulatory network. (c) The coefficient and biases can be used to predict the expression levels of genes from the expressions of TFs in a new sample. The difference between the predicted and actual expression levels is calculated for each set of models and used to determine the DRS score. The DRS score determines the predicted class label $\hat{c}$ of the new sample.

and METABRIC and ACES datasets have 12,342 genes and 998 TFs in common. We selected these datasets as they are among the most widely used benchmarks for evaluating breast cancer classification performance.

C. Construction of Metastatic and Non-Metastatic GRNs

A gene regulatory network can be denoted as the ordered pair (V, E) where V is the set of genes, and E is the set of directed edges. Given the gene set V , the network inference essentially boils down to the inference of E from gene expression profiles. The network construction method employed in this study relies on the underlying assumption that the expression level of a gene can be approximated as an affine function of the expression levels of its regulatory TFs within a specific tissue or disease context. Let S denote the set of samples that belong to a particular context. The observed expression levels of the gene j in all samples from the set S can be modeled as a regression problem, where the TF expression levels are independent variables (1), (2).

$$y_{k,j} \simeq b_j + \mathbf{X}_{k,:} \mathbf{a}_j \quad \text{for } k \in S \quad (1)$$

$$\text{where } b_j \in \mathbb{R}, \mathbf{a}_j = [a_1, \dots, a_t]^\top \in \mathbb{R}^{t \times 1} \quad (2)$$

The observed variables are y and $\mathbf{X}$, and the parameters denoted $b, \mathbf{a}$ are to be estimated. The scalar b is the bias. The magnitude of a_i can be interpreted as the influence of the i -th TF on the expression of the target gene, and the sign indicates the direction of regulation (e.g. up or down-regulated). This model can be trivially transformed into a partial GRN by connecting gene j to every TF i with an edge such that $|a_i| > 0$. Given the large number of TFs compared to the limited number of samples, overfitting can occur. We used LASSO regression to penalize the number of regulatory TFs per gene and obtain sparse feature

coefficients, as most genes are expected to be regulated by only a small subset of TFs. Given the high dimensionality of gene expression datasets, LASSO models can also be effective in mitigating the effect of noise present in the training data. LASSO regression requires tuning a single hyperparameter λ which in our case offers a trade-off between sparsity and the goodness-of-fit measure, R^2 . We tuned λ by inspecting the empirical distribution of R^2 for geometrically increasing values of λ (Supplementary Figure S1). Training the regression models for each gene yields $\mathbf{a}_j$ for every gene j , which can be transformed into partial GRNs followed by consolidation into a genome-wide regulatory network.

We applied the network construction strategy outlined above to acquire two GRNs from the two disjoint sets of samples S^+ and S^- which correspond to metastasis and non-metastasis, respectively. For global network construction, we excluded the genes with $R^2 \leq 0.1$, using 10-fold cross-validation on all patient samples, as these genes did not fit the model well. The set of remaining genes is V' , where $|V'| = n = 5,910$, which is approximately 46% of all genes in the ACES dataset. As we will observe later, this step does not affect classification performance. After the gene removal step, 1,161 bootstrapped samples from each outcome group were drawn 200 times to train lasso regression models for gene expression using the scikit-learn library [34], with oversampling applied to S^+ . We used $\lambda = 0.06 \simeq 2^{-4}$ to enforce sparsity while achieving moderately high R^2 . The mean coefficients from the models were used to construct the final model for each gene. We denote $(b_j^+, \mathbf{a}_j^+)$ and $(b_j^-, \mathbf{a}_j^-)$ as the bootstrap-mean bias and coefficients for the expression levels of the j -th gene in metastatic and non-metastatic samples, respectively. The bias terms are collectively denoted as row vectors $\mathbf{b}^+ = [b_1^+, \dots, b_n^+]$ and $\mathbf{b}^- = [b_1^-, \dots, b_n^-]$ and were used in later sections for classification. The TF-gene coefficients

for all genes in V' within a metastatic context were learned from the observed expression levels $\mathbf{Y}_{k,:}$ and $\mathbf{X}_{k,:}$ where $k \in S^+$. The coefficients are collectively denoted as the matrix $\mathbf{A}^+ = \{a_{ij}\} = [\mathbf{a}_1^+, \dots, \mathbf{a}_n^+] \in \mathbb{R}^{t \times n}$. As an additional denoising step, we applied a cut-off value of $\alpha = 0.02$ on the absolute value of the coefficients in $\mathbf{A}^+$. We then used $\mathbf{A}^+$ to acquire the directed GRN for metastatic samples as follows: $G^+ = (V', E)$ where $E = \{(i, j), i \in T, j \in V', |a_{ij}| > 0\}$. Applying the identical procedure on S^- allows us to acquire $\mathbf{A}^-$ and consequently G^- . The model collections $\mathbf{A}^+$, $\mathbf{A}^-$ include models for the TFs as well. Each TF's expression was zeroed in the input expression when learning the model for itself.

D. Network Analysis and Comparison

Network statistics including the number of singletons, size of the largest connected component, the mean and maximum degree, the mean shortest distance, diameter, as well as the clustering coefficient, were measured using the NetworkX python package [35] for the networks G^+ and G^- . The clustering coefficient was calculated on the undirected versions of the networks. The mean shortest distance and the diameter was calculated for the largest connected component only. The connectivity difference of each gene between the two networks was calculated as the Euclidean distance between the respective connection vectors [36]. Genes with high connectivity differences were considered to be highly rewired genes. For every pair of TFs with no fewer than 10 targets in both networks, the correlation between the two respective connection vectors was calculated for each context. KEGG pathway [37] enrichment analysis on the 200 most-rewired genes was performed using ShinyGO [38] with an FDR cutoff of 0.05 and pathways containing between 15 and 1000 genes. The ranked list of rewired genes was analyzed for enrichment in the COSMIC CGC genes [39] using the GSEA software package [40] with the default parameter settings.

E. Construction of TF-TF Co-Regulatory Network

Changes in systems-level regulatory coordination between the two prognostic outcomes can be observed on the scale of transcription factors by summarizing the GRNs into TF-TF co-regulatory networks. The TF-TF networks contain mostly hub TFs with many targets and are built around the idea of positive and negative co-regulation. If two TFs u, v regulate a gene j such that $a_{uj}a_{vj} > 0$, we consider j to be positively co-regulated by u, v . Conversely, $a_{uj}a_{vj} < 0$ indicates negative co-regulation. Formally, we use the following variants of the step function that can be used to map a regression coefficient to a network adjacency:

$$\llbracket a \rrbracket = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \llbracket -a \rrbracket = 1 - \llbracket a \rrbracket \quad (3)$$

Without loss of generality, these functions can be applied to matrices. We acquired the $t \times t$ adjacency matrices for the metastatic TF-TF positive co-regulation and negative co-regulation networks, H_{pos}^+ and H_{neg}^+ , as follows:

$$H_{pos}^+ = \llbracket \llbracket \mathbf{A}^+ \rrbracket \llbracket \mathbf{A}^+ \rrbracket^\top + \llbracket \mathbf{A}^+ \rrbracket \llbracket \mathbf{A}^+ \rrbracket^\top - \Delta \rrbracket \quad (4)$$

$$H_{neg}^+ = \llbracket \llbracket \mathbf{A}^+ \rrbracket \llbracket \mathbf{A}^+ \rrbracket^\top + \llbracket \mathbf{A}^+ \rrbracket \llbracket \mathbf{A}^+ \rrbracket^\top - \Delta \rrbracket \quad (5)$$

H_{pos}^+ contains an edge between two TFs only if they share at least Δ positively co-regulated targets. Similarly, H_{neg}^+ connects two TF with an edge only if they share at least Δ negatively co-regulated targets. For this analysis, we used $\Delta = 100$, which resulted in small, sparse networks containing mostly hub TFs. Finally, we combined H_{pos}^+ and H_{neg}^+ to acquire the adjacency matrix of the unified TF-TF co-regulatory network for metastatic patients, $H^+ = H_{pos}^+ + 2H_{neg}^+$. Every entry in H^+ is an integer in the range $[0, 3]$, where 0 denotes no co-regulation, 1 denotes positive co-regulation, 2 denotes negative co-regulation and 3 denotes both positive and negative co-regulation. The same process was used to build the TF-TF co-regulatory network H^- from $\mathbf{A}^-$.

F. Dysregulation Score as a Simple Classifier for Predicting Breast Cancer Metastasis

Metastatic cells exhibit structural and biochemical differences from non-metastatic cells [41] through the rewiring of gene networks and changes in gene expression patterns. These relative differences are incorporated into the regression models, as each model is built from a specific class of samples. Using the two sets of models, we can generate two estimates of the expression levels of each gene for a previously unseen sample with an unknown label. One of these estimates can be expected to be more concordant with the ground truth expression levels, which can be exploited to classify the sample.

To evaluate the classification performance of this approach, we split the samples into training and test sets. The samples in the training data were split into two subsets S^+ and S^- again, and the observed expression levels in each subset were used to estimate the LASSO coefficients. Given a sample $\mathbf{y} = [y_1, \dots, y_n]$ from the test set, we can extract the expressions of the TFs, $\mathbf{x} = [y_i : i \in T]$. For some set of regression coefficients $\mathbf{A}$ and $\mathbf{b}$, the predicted expression of all genes in that sample is $\hat{\mathbf{y}} = \mathbf{x}\mathbf{A} + \mathbf{b}$. Since we have two different sets of coefficients trained from S^+ and S^- , we can calculate the distance between the predicted and actual expressions for each set as follows:

$$d^+(\mathbf{y}) = \text{distance}(\mathbf{y}, \mathbf{x}\mathbf{A}^+ + \mathbf{b}^+) \quad (6)$$

$$d^-(\mathbf{y}) = \text{distance}(\mathbf{y}, \mathbf{x}\mathbf{A}^- + \mathbf{b}^-) \quad (7)$$

We used Pearson distance as the distance function, as it is invariant to linear transformations commonly applied to gene expression data during preprocessing or normalization. The dysregulation score of a sample is defined as the difference between the two distances:

$$DRS = d^-(\mathbf{y}) - d^+(\mathbf{y}), \quad (8)$$

and the predicted binary class label $\hat{c}$ can be obtained using the DRS score. If a sample from a test set has a true label c and $c = \hat{c}$, it is classified correctly.

The classification performance of DRS was compared with several supervised machine learning algorithms, including Random Forest (RF), Multilayer Perceptron (MLP), Logistic Regression (LogR), Decision Tree (DecTree), k -nearest-neighbor

(KNN), and Support Vector Classifier (SVC) implemented in the python scikit-learn library [34]. For SVC, a linear kernel was used. All other hyperparameters were set to the default values used by sklearn 1.2.2. The feature set used for these models was the expression of all genes. To account for inter-cohort differences, we conducted stratified 10-fold cross-validation and a leave-one-study-out cross-validation on the complete dataset. Given the class imbalance, we ensured a fair evaluation of the models using multiple performance metrics: Area Under ROC (AUC), kappa statistic, F1 score, and balanced accuracy. The metrics were calculated from the number of true-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) prediction outcomes as follows: $Precision = \frac{TP}{TP+FP}$; $Recall = \frac{TP}{TP+FN}$; $TruePositiveRate = Sensitivity = \frac{TP}{TP+FN}$; $FalsePositiveRate = \frac{FP}{TN+FP}$; $Specificity = \frac{TN}{TN+FP}$; $F1\text{-score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$; $Balanced\text{-accuracy} = \frac{(Sensitivity + Specificity)}{2}$; and $Kappa = \frac{2(TP \times TN - FN \times FP)}{(TP+FP)(FP+TN) + (TP+FN)(FN+TN)}$. The decision threshold for each model including DRS was set at the midpoint of the range of probability scores produced by that model for all test cases. Determining the threshold for individual models instead of using a default threshold of 0.5 led to more robust performance for models such as RF and SVC.

We also compared the performance of the models on the NKI and METABRIC datasets in two different settings. In the first setting, models were trained using 10-fold cross-validation to evaluate performance on the secondary datasets. In the second setting, the additional datasets were quantile normalized to match the distribution of the ACES dataset. For this experiment, all models were retrained on the ACES dataset using only the genes that were available in the secondary datasets. The NKI and METABRIC datasets were then used as test sets to evaluate performance.

G. Reproducibility

All randomized algorithms were initialized with a fixed random seed to ensure reproducibility. The source code and data files are available on GitHub¹.

III. RESULTS

A. Regulatory Rewiring of Cancer-Related Genes

The two gene regulatory networks G^+ and G^- vary on a number of topological properties. While both networks are explicitly made to have the same number of target genes, G^+ has more edges and non-isolate TFs than its non-metastatic counterpart (Table I). It has a higher mean in-degree, representing more regulators per target gene and a generally more complex regulatory network. Both networks' in- and out-degree distributions are highly right-skewed, with a much larger mean degree than the median, which is a common characteristic of scale-free networks. The largest out-degree of G^+ is significantly smaller compared to G^- , suggesting that the regulatory connections

TABLE I
METASTATIC (G^+) AND NON-METASTATIC (G^-) NETWORK CHARACTERISTICS.

Network property	G^+	G^-
TFs	505	455
Targets	5,910	5,910
Total nodes	5,992	5,969
Edges	42,590	34,723
Largest out-degree (TF)	1,590	2,076
Largest in-degree (Target)	97	104
Mean/median out-degree (TF)	84.34/20	76.31/16
Mean/median in-degree (Target)	7.21/5	5.88/4
Clustering coefficient	0.22	0.26
Nodes in largest connected component	5992	5960
Mean shortest distance ¹	2.82	2.86
Diameter ¹	8	8

¹calculated for the largest connected component

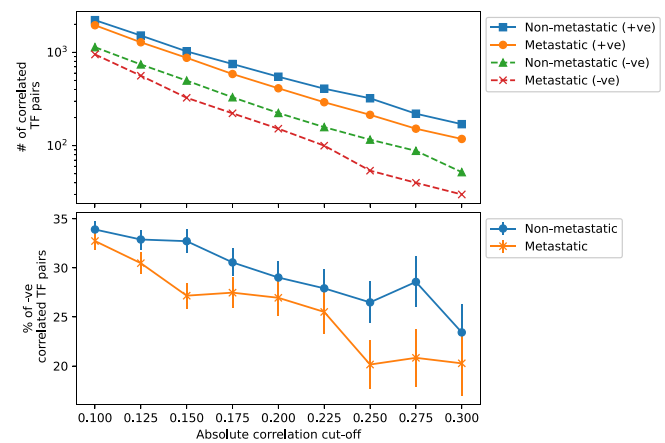


Fig. 2. Number of +ve and -ve TF pairs in metastatic and non-metastatic contexts (top), and the percent of -ve pairs in each context (bottom). The error bars in the bottom panel represent one standard deviation of the measurement estimated with binomial distribution. TFs with at least 10 targets in both networks are considered.

in G^+ are more spread out. Notably, G^+ has a much lower clustering coefficient.

The correlation between every pair of TF connection vectors, calculated for the two disease contexts, paints an interesting picture. We categorized every TF pair as a +ve pair or -ve pair based on the sign of the correlation coefficient and for each context, inspected the number of pairs with absolute correlation greater than or equal to a threshold (Fig. 2). For any given cut-off correlation value, the total number of +ve and -ve pairs is lower in the metastatic context compared to its counterpart. In contrast, the percentage of -ve pairs is significantly lower in the metastatic context.

Individual genes exhibiting high rewiring between the two GRNs were revealed to be mostly cancer-associated genes. The 30 highest-rewired target genes and TFs (Table II) include metastatic breast cancer marker genes such as SCGB2A2 [42], SCGB1D2 [42], TFF1 [43], as well as migration promoters and stimulators KRT14 [44] and CECAM6 [45]. In a recent study, it has been shown that DHRS2 inhibits the growth and metastasis in ovarian cancer [46]. As expected, ESR1 is the most rewired TF, consistent with its paramount importance in breast cancer

¹<https://github.com/Tanzira/DRS>

TABLE II
MOST-REWIRED NETWORK NODES IN DECREASING ORDER OF CONNECTIVITY DIFFERENCE.

Genes	TFs
SCGB2A2, SCGB1D2, DHRS2, SCGB2A1, S100A7, CPB1, CXCL14, MMP1, CXCL13, CA2, CEACAM6, S100P, CALML5, AQP3, TFF1, SLC1A1, SCUBE2, KRT15, PIP, NPY1R, KRT14, COL2A1, PEG10, TFF3, CRABP1, APOD, SPP1, VTCN1, CSTA, GJA1	ESR1, GATA3, NFIB, MYBL1, LTF, TFAP2B, POU2AF1, ZBTB18, MYB, SOX10, FOXA1, ZEB1, MYC, ELF5, FOS, TCF4, MAFF, SOX11, EN1, SPDEF, SOX9, NFIL3, TRPS1, DACH1, NR2F1, EHF, HOXC10, FOXM1, SNAI2, EGR1

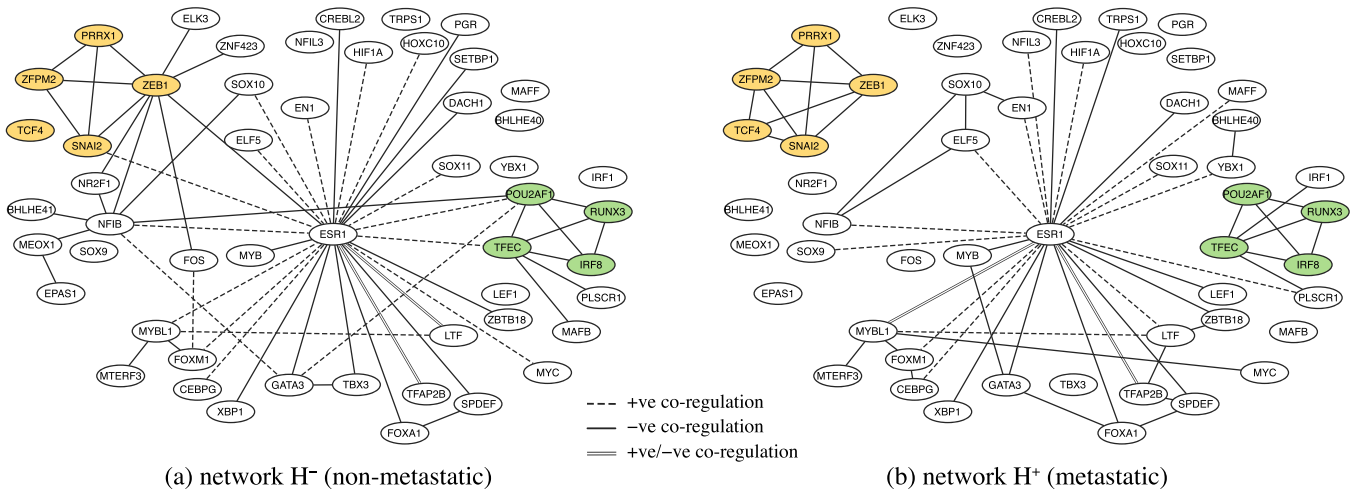


Fig. 3. TF-TF co-regulatory networks. (a) network H^- (non-metastatic) has more edges, with the EMT and immune response TFs having more connections. (b) network H^+ (metastatic) has fewer edges, with the EMT and immune response TFs notably disconnected.

development and progression [47]. Other TFs with high connectivity differences play important roles as makers or components of metastasis, such as TFAP2B [48], FOS [49], GATA3 [50], ELF5 [51], NFIB [52], and MYBL1 [53]. ZBTB18 is associated with brain tumor [54], and more recently, has been identified as a transcriptional repressor that prevents metastasis through reduction of chromatin accessibility in a breast cancer context [55]. LTF, another highly rewired TF, is involved in innate immune responses. Several studies revealed that upregulation of LTF is significantly associated with metastasis-free survival in prostate cancer and Clear Cell Renal Cell Carcinoma [56], [57]. A comprehensive list of rewired TFs with the rewiring score is included in Supplementary Table S1. The 200 most-rewired genes were enriched in multiple KEGG pathways related to cancer (Supplementary Figure S2a). Gene set enrichment analysis revealed that highly rewired genes were enriched in the COSMIC CGC list of cancer-affiliated genes (Supplementary Figure S2b). The data-driven transcriptional regulation inference and the subsequent quantitative comparison accurately implicate many highly-rewired genes in cancer development and growth.

B. Characterization of TF-TF Co-Regulation

The TF-TF co-regulatory networks H^+ and H^- built around hub TFs reveal subtle but important rewiring between metastatic and non-metastatic samples. The two networks share an almost identical set of non-isolate TFs and the majority of edges, with ESR1 as the hub surrounded by key TFs playing essential roles in breast cancer development and progression, such as ZEB1,

SNAI1, MYB, FOXA1, and GATA3 (Fig. 3). One of the most salient differences between the two networks involves a group of TFs regulating the epithelial-mesenchymal transition (EMT) of cells, a hallmark event of metastasis [58]. ZEB1, a master regulator of EMT, forms a group with other EMT-inducing genes SNAI2, ZFP22, and PRRX1 in both co-regulatory networks, consistent with previous findings [59]. The EMT group appears connected to the central hub in the network H^- , whereas it is disconnected from the rest of the network in H^+ . The two networks include another common subnetwork of four TFs (TFEC, IRF8, RUNX3, POU2AF1) that are markers of immune cell subtypes identified from nasopharyngeal carcinoma tumor microenvironment [60], where its connectivity to ESR1 and other TFs is significantly reduced in the metastatic network.

Other notable TFs with reduced co-regulation include FOS, NFIB, NR2F1, and GATA3. FOS is a sub-unit of the AP-1 transcription factor, a master regulator of cell proliferation and apoptosis [61], NFIB is a master regulator of cell differentiation and a crucial driver of cancer metastasis, shown to promote metastasis through a widespread increase in chromatin accessibility [62]. NR2F1 is a tumor dormancy biomarker and plays a critical role in inducing and maintaining dormant disseminated tumor cells, which is believed to be the origin of metastases [63]. ESR1, FOXA1, and GATA3 interact to form an enhanceosome to regulate ER-induced cell growth in ER+ breast cancer cell lines [64]. Recent studies show a significantly increased binding of FOXA1 and widespread epigenetic reprogramming in ESR1 enhancer sites in drug-resistance cancer cell lines and patients with poor outcomes [65], [66]. Our results suggest that the

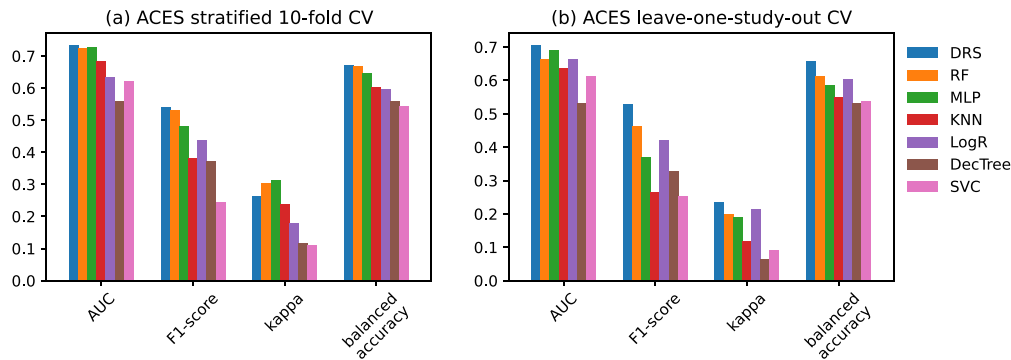


Fig. 4. Performance comparison of *DRS* with other classifiers on the ACES dataset. (a) performance metric scores in a 10-fold cross-validation setting. (b) performance comparison in a leave-one-study-out cross-validation setting. *DRS*: DysRegScore-based classifier. RF: Random Forest. MLP: Multilayer Perceptron. LogisticR: Logistic Regression, DecTree: Decision Tree. KNN: K-nearest-neighbor Classifier. SVC: Support Vector Classifier.

functional consequences of the FOXA1-mediated epigenetic reprogramming may be the loss of co-regulation between the cancer cell EMT (ZEB1, SNAI2), proliferation and apoptosis (AP-1), differentiation (NF1B), dormancy (NR2F1), and immune response (POU2AF1) transcriptional programs. The loss of *co-regulation* resulted from dysregulation, as opposed to the loss of regulation of these otherwise dynamic processes, may be the driving force for metastasis.

C. Performance of Dysregulation-Based Breast Cancer Metastasis Prediction

We defined a dysregulation score based on the relative errors that the two sets of regression models made in recapitulating the expression levels of target genes in a previously unseen patient sample. Intuitively, the set of models more closely estimating the ground-truth expression levels is likely fitted from the training patients having the same disease context as the new patient. It should be emphasized that the regression models themselves only predict target gene expression based on the expression levels of TFs, and are not trained on the patient class labels. The classification function is therefore only a byproduct of the GRN inference and requires no additional learning or parameter estimation.

The performance of this simple distance-based classifier, DysRegScore (*DRS*), was compared with several supervised learning algorithms and evaluated on multiple metrics using cross-validation. *DRS* has the highest AUC, F1-score, and balanced accuracy for the ACES dataset in a stratified 10-fold cross-validation setting. *DRS*, Random Forest, and Multilayer Perceptron have relatively higher AUC scores of 0.735, 0.724, and 0.728, respectively (Fig. 4(a)). The models were also tested on each of the twelve patient cohorts in the ACES dataset after training on the remaining eleven cohorts for the *leave-one-study-out* (*LOSO*) performance analysis. This was a more challenging evaluation than 10-fold cross-validation, as the molecular and clinical characteristics of patients in the testing cohort can differ significantly from the training cohorts. In this setting, *DRS* achieved the highest AUC, F1, kappa, and balanced accuracy (Fig. 4(b)). In addition, we used two existing GRN inference methods, TIGRESS [67] and Bayesian Ridge Score, from the

GRNaDIne python package [68] as preprocessing steps for our method. The GRNs inferred using these methods were used to preselect a subset of the TFs to train LASSO regression models, which resulted in a slight decrease in prediction performance (Supplementary Table S2).

To test the reliability and robustness of our *DRS* classifier, we used two independent cohorts, NKI and METABRIC. Baseline models alongside *DRS* were evaluated using the validation cohorts in a 10-fold cross-validation setting using the same λ that was used for the ACES dataset. The evaluation results indicate that *DRS* is the best-performing classifier, with the highest scores in nearly every metric (Fig. 5(a)–(b)). In addition, the AUC increased from 0.715 to 0.736 when the parameter λ was changed to 0.03 (Supplementary Table S3). We also used these cohorts as test sets for models trained on the ACES dataset after gene identifier mapping. *DRS*, along with other state-of-the-art models, were trained on a subset of gene expressions in the ACES dataset using genes and TFs for which expression levels are also available in the validation cohorts. The validation datasets were quantile normalized to match the distribution of the ACES expression levels. *DRS* outperforms other baselines in almost all metrics (Fig. 5(c)–(d)). *DRS* has the second highest AUC of 0.671 in METABRIC with KNN having the highest AUC of 0.677. For the NKI data set, *DRS* has the second highest Kappa value of 0.245 with Random Forest achieving a Kappa score of 0.250. *DRS* retains its position as the classifier with the highest F1 score and balanced accuracy. Statistical significance was evaluated by measuring performance metrics for each test split and comparing the difference between *DRS* and the baseline classifiers. *DRS* outperforms the baselines in 84 of 96 pairwise comparisons (performance metric difference ≥ 0), 48 of which are significant with $p \leq 0.05$ (Supplementary Table S4).

We tested whether parameter tuning can drastically alter the observed classification performance. *DRS* exhibits stable AUC scores on the ACES dataset in both 10-fold and *LOSO* cross-validation for multiple different configurations of the regularization parameter λ , denoising using R^2 -based gene selection, and coefficient cut-off α (Table III). As an example, even if the R^2 -based gene selection and coefficient-based edge selection steps (see Methods) are omitted while $\lambda = 0.06$ is unchanged, the AUC dropped only slightly from 0.735 to 0.731. The

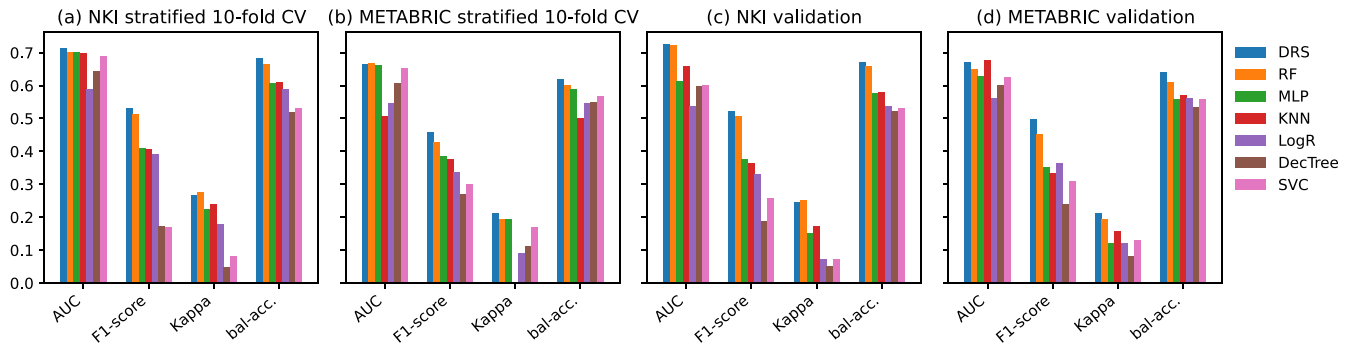


Fig. 5. Performance comparison of DRS and other classifiers on the validation datasets. (a) Classifiers were trained on the NKI dataset and tested using stratified 10-fold cross-validation. (b) Classifiers were trained on the ACES dataset, and tested on the quantile-normalized NKI dataset. (c) Classifiers were trained on the METABRIC dataset and tested using stratified 10-fold cross-validation. (d) Classifiers were trained on the ACES dataset, and tested on the quantile-normalized METABRIC dataset.

TABLE III
EFFECT OF HYPERPARAMETER TUNING ON CLASSIFICATION PERFORMANCE (AUC) OF DRS ON THE ACES DATASET.

R^2 cutoff	Hyperparameter			# genes	AUC	
	λ	α			10F	LOSO
0.1	0.13 ($\approx 2^{-3}$)	0.02	2,927	0.720	0.705	
0.1	0.06 ($\approx 2^{-4}$)	0.02	5,910	0.735	0.707	
0.1	0.03 ($\approx 2^{-5}$)	0.02	8,911	0.732	0.689	
None	0.13 ($\approx 2^{-3}$)	0	12,750	0.717	0.696	
None	0.06 ($\approx 2^{-4}$)	0	12,750	0.731	0.695	
None	0.03 ($\approx 2^{-5}$)	0	12,750	0.725	0.675	
None	[0.13, 0.06, 0.03] [†]	0.02	12,750	0.734	0.697	

[†]model ensemble

denoising significantly reduces weakly correlated TF-gene interactions which helps to understand dysregulation at the network level. The classifier performance is not sensitive to the choice of these hyperparameters. We also trained additional baselines using a gene-filtering method identical to the R^2 -based filtering in our pipeline. We observed that filtering slightly reduces the performance of all baseline methods except RF, for which the performance increases only slightly (Supplementary Table S5). This suggests that gene selection alone cannot account for the predictive performance of DRS.

IV. DISCUSSION

Comparative analysis of the topological properties of the two gene regulatory networks G^+ and G^- indicates distinctive network-level features associated with the corresponding metastatic and non-metastatic phenotypes (Table I). It is evident from the network level properties that G^- has a more modular organization and regulatory hierarchy that is disrupted in its metastasis-associated counterpart. This is further supported by the fact that many genes and TFs with the highest rewiring scores are implicated in cancer and cell migration in prior studies (as shown in Section III-A) [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57].

Interestingly, analysis of the co-regulatory networks revealed that, while the metastatic network contains fewer TF pairs that co-regulate their targets, this reduction is more pronounced for TFs with negative co-regulations than for those with positive

co-regulations (Fig. 2). As most transcriptional regulation is achieved via combinatorial binding to DNA sequences [69], the sign of the correlation between TF pairs may reflect how the binding of one TF affects the binding of the other TF on the target gene promoters and therefore is closely related to binding cooperativity, a concept extensively studied in biochemistry [70]. Negative cooperation, in which the binding of one regulator decreases the binding affinity of the other, in a complex system is important in determining the system's response behavior to stimulus [71]. In general, negative cooperation makes a multimeric regulator's response more graded; however, if the concentration of the regulator is limited, the negative cooperativity may result in an ultrasensitive response with a pronounced threshold [72]. Therefore, the different ratio of +ve to -ve pairs between the two cancer co-regulatory networks may very well reflect a dramatically changed state of the systems-level regulatory landscape in metastatic cells that are important for their unique life cycles. This interesting hypothesis can be further investigated with additional data.

Differences between the connectivity of hub TFs in the TF-TF co-regulatory networks H^+ and H^- highlight regulatory rewiring that plays a crucial role in metastasis (Fig. 3). The central-hub organization of the TF-TF co-regulatory networks H^+ and H^- is consistent with recent studies that show the existence of a cluster of TFs interacting with ESR1 to form a so-called megaTrans complex and bind to ERE-containing enhancers to modulate a broad transcriptional program in breast cancer cell lines [66], [73]. The EMT group is connected to the hub in the metastasis-associated network H^- , whereas it is an isolated module in H^+ . Although many studies have tried to determine whether EMT expression may be a prognostic factor, the results are often inconclusive. A more recent large-scale meta-analysis showed that ZEB1 and most EMT TFs are not strongly associated with survival, while the upregulation of SNAI2 is only modestly associated with survival [74]. Our analysis suggests that the loss of connectivity between ESR1 and the groups associated with EMT and immune response is much more important in the context of metastasis than changes in individual expression levels of the member genes.

The distinct regulatory connections learned from each disease context can be readily applied to metastasis prediction with no

additional model training or tuning. DRS outperforms baseline models in most experimental settings, including 10-fold cross validation and secondary validation datasets for a number of different metrics (Figs. 4–5). Compared to when coupled with external GRN inference methods such as TIGRESS as feature selectors, vanilla DRS performs better. This strongly suggests that the LASSO regression models that underpin the DRS classifier are sufficiently capable of selecting robust features, which translate to superior predictive performance. The observed performance is stable across different hyperparameter settings and model configurations, and our approach performs surprisingly well alongside supervised models such as Random Forest which is generally well-suited to tabular data. The performance of the DRS classifier can be attributed to its foundation in GRN-driven biological insights that are consistent across different cohorts.

To the best of our knowledge, our method is the first direct application of knowledge models for accurate classification, bridging a significant gap in current literature. This framework can be extended in several ways. For example, more sophisticated gene expression regressors can be explored as potential base models for GRN-building. Ground-truth biological networks can also be used as priors during the GRN-building phase with associated tuning parameters. As GRNs learned from different patient cohorts can be expected to exhibit some differences, GRNs inferred from one dataset can be used to fine-tune the GRNs learned from other datasets. A broad investigation of how hyperparameter tuning can affect the performance can provide useful insights. The applicability of this method in other biological contexts within and beyond the scope of breast cancer can be investigated in future studies.

V. CONCLUSION

In this work, we developed a computational framework to simultaneously achieve knowledge discovery and knowledge-based, explainable prediction. Applying it to study breast cancer metastasis, we utilized gene regulatory networks (GRNs) as knowledge models and constructed separate GRNs for metastatic and non-metastatic patients. We demonstrated that these networks effectively represent the internal cellular states corresponding to their respective phenotypes and can be directly employed for classification purposes. These networks not only helped identify highly rewired genes that are strongly associated with metastasis and enriched in cancer-relevant pathways, but also revealed that the loss of co-regulation among the master regulators of several biological pathways is a key feature of metastasis, suggesting potential therapeutic strategies. A simple-to-explain scoring system based on the GRNs achieved remarkable prediction accuracy when compared to state-of-the-art classifiers, indicating that the knowledge models are valid and powerful predictive tools. It is also worth noting that many sophisticated GRN construction methods are available, which may further enhance the quality of the knowledge models and the accuracy of knowledge-based predictions. Our work bridges a significant gap between knowledge discovery and data-driven phenotype prediction, as demonstrated in the context of breast

cancer metastasis, and has the potential for application to other complex diseases.

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